Reducing 30-Day Readmissions & Revenue Leakage at Sunrise Multispeciality Hospital

Problem statement, dataset guide and deliverables - Dashboard | Predictive Model | GenAI RAG Chatbot (Gemini)

1. Business Context (real-world grounding)

- Indian private hospitals report 30-day readmission rates of roughly **8-14%**, rising toward ~18% for chronic patients without structured post-discharge follow-up.
- Large listed chains run at **~64-68% bed occupancy**; growth now depends on using existing beds better (ALOS, occupancy, revenue per bed).
- Government schemes (AB-PMJAY, ~38,000 empanelled hospitals) settle claims within a 15-day turnaround target, and AI-based fraud/claim checks are tightening - documentation errors now directly cost revenue.
- Hospitals are expected by NABH and payers to show quality metrics: readmission, mortality, LAMA, ER wait, patient experience.

2. The Scenario

Sunrise Multispeciality Hospital is a **fictional 300-bed NABH-accredited hospital in Coimbatore** serving Coimbatore, Tiruppur, Erode, Pollachi, Palakkad and Salem. FY2025-26 (Apr-2025 to Mar-2026) closed with:

KPI	FY25-26 actual	Target
30-day readmission rate	12.6%	< 10%
Bed occupancy (overall)	63.1%	75-80%
Average length of stay	6.24 days	< 5.8 days
Insurance claim rejection	9.2% (INR 9.03 Cr)	< 5%
48-hr follow-up call coverage	54.8%	> 90%
Net Promoter Score	6.4	> 40

The CEO has observed that Cardiology, Nephrology and Neurology are running near full capacity (85-88% occupancy) while OBG and General Medicine are under-utilised, readmissions are concentrated in chronic-disease patients, and insurance rejections are eroding margin.

3. Problem Statement

As a Healthcare Data Analyst / AI Engineer, use FY2025-26 hospital data to (a) explain what drives 30-day readmissions, bed bottlenecks and claim rejections, (b) predict which patients are at high risk of readmission at the time of discharge, and (c) give doctors, nurses and administrators an AI assistant that answers questions from the hospital's policy manual and annual report.

4. Deliverables

#	Deliverable	Tools	Key outputs
1	Executive Dashboard	Power BI / Tableau / Looker Studio / Streamlit	KPI cards, monthly trend, dept heat-map, payer mix, claim rejection Pareto, NPS
2	Readmission Risk Model	Python - scikit-learn / XGBoost	Binary classifier on *readmitted_30d*; ROC-AUC, recall at top-20% risk, SHAP drivers, risk-band list

#	Deliverable	Tools	Key outputs
3	RAG Chatbot	Gemini API (gemini embeddings + Gemini Flash), LangChain/LlamaIndex, ChromaDB/FAISS, Streamlit	Answers grounded in Policy Manual + Annual Report with page citations
4	Recommendation Memo	Slides / doc	3-5 actions with estimated INR impact

5. Dataset Files (synthetic, FY2025-26)

File	Rows	Grain	Use it for
admissions.csv	11,200	1 row per inpatient admission	Core fact table: dashboard + ML target *readmitted_30d*
patients.csv	6,500	1 row per patient	Demographics, comorbidities, payer (dimension)
departments.csv	9	1 row per department	Beds, ICU beds, consultants (dimension)
insurance_claims.csv	~6,900	1 row per insured admission	Claim status, rejection reason, TAT, revenue leakage
daily_bed_census.csv	3,285	department x day	Occupancy trend, ICU load, admissions/discharges flow
patient_feedback.csv	3,200	1 row per surveyed discharge	Ratings, NPS, free-text comments
Sunrise_Hospital_Policy_Manual.pdf	-	RAG document	Chatbot knowledge base: SOPs, timelines, rules
Sunrise_Annual_Report_FY2025-26.pdf	-	RAG document	Chatbot knowledge base: performance, finance, strategy

Joins: admissions.patient_id -> patients | admissions.dept_id -> departments | insurance_claims.admission_id / patient_feedback.admission_id -> admissions | daily_bed_census.dept_id -> departments

6. Data Dictionary - admissions.csv (main table)

Column	Description
admission_id, patient_id	Unique keys
age, gender, city, distance_km	Patient demographics; distance from hospital
dept_id, department, icd10_code, diagnosis	Treating department and primary ICD-10 diagnosis
admission_type	Emergency / Elective
admit_date, discharge_date, los_days	Dates and length of stay (days)
icu_stay	TRUE if any ICU stay
er_wait_minutes	ER door-to-admission wait (Emergency only; few nulls - clean it)
attending_doctor	Masked doctor code
payer	Cash, Private Insurance, PMJAY, CMCHIS, ESI, Corporate
comorbidity_count, diabetes, hypertension, ckd, copd, heart_disease, smoker, bmi	Clinical risk factors (bmi has a few nulls)
prior_admissions_12m	Earlier admissions of same patient in the year
discharge_disposition	Discharged Home / LAMA / Transfer Out / Death
discharge_hour	Hour of discharge (24h)
med_reconciliation_done, followup_call_48h, followup_visit_7d	Care-transition process flags
total_bill_inr	Final bill amount
readmitted_30d	**TARGET** - readmitted within 30 days (eligible: Home/LAMA)
days_to_readmission	Days until readmission (leakage - drop for modelling)

7. Suggested Analysis Questions

- Which departments and diagnoses drive readmissions? How does 48-hr follow-up change the rate?
- Where are the bed bottlenecks by month and department? What ALOS reduction frees how many bed-days?
- Which payer and rejection reason causes the most revenue leakage (Pareto)?

- Does ER wait time or readmission reduce patient ratings / NPS?

8. Modelling Guidelines

- Filter to discharge_disposition in (Discharged Home, LAMA). Drop IDs, dates, days_to_readmission (leakage).
- Class imbalance ~12.6% positive - use stratified split, class_weight, evaluate ROC-AUC + PR-AUC + recall@top-20%.
- Baseline: Logistic Regression; then Random Forest / XGBoost. A reasonable target is ROC-AUC 0.72-0.80.
- Explain with SHAP; convert scores into Low / Medium / High risk bands for the discharge team.

9. RAG Chatbot - Sample Test Questions

- What is the target time from discharge order to patient exit?
- Who must be called within 48 hours of discharge and what is checked?
- Within how many days must an insurance claim be submitted after discharge?
- What was the hospital's readmission rate in FY2025-26 and which department was highest?
- What actions are planned for FY2026-27 to reduce claim rejections?
- What is the escalation trigger when ICU occupancy crosses the threshold?

All data, names and documents are synthetic. Market context drawn from public reporting (Economic Survey of India, Statista/industry reports, PIB/Parliament replies on AB-PMJAY, 2025-2026).